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Foundations of Python
Assignment 06 – documentation
https://github.com/SteveRowse/Assignment-06

Summary of module 06

In Module 06 we covered the following topics:

- Functions
  - Parameters
  - arguments

  Global vs local functions
  Overload Functions
  Classes
  Separation of Concerns

Steps to complete assignment 06

The Assignment:
Create a Python program that demonstrates using constants, variables, and print statements to display a message about a student's registration for a Python course. This program is very similar to Assignment05, but It adds the use of functions, classes, and using the separation of concerns pattern.

starting from the starter file I:

1. began by changing the header
2. making my two classes: FileProcessing and IO or Input Output. I also separated these with a #---- just to visually see the two sections.
3. Put in my class headers

```
# Processing ---------------------------------------- #
class FileProcessor:
#------------------------------------------------------------------

# Presentation ---------------------------------------- #
class IO:
```

Figure 1: outlining the classes

4. Started parsing out sections of code, making empty functions under the Classes.

```python
# Processing ------------------------------------- #
class FileProcessor:
    @staticmethod
    def read_data_from_file():
    pass
        """ This function reads the .json file into memory
            ChangeLog: (Who, When, What)
            SRowse,3.4.26,Created function
            :return: Student data
        """
```

Figure 2: blocking in Functions and headers

5. I cut code from the main body and pasted the code under the functions and assigned new variables to the parameters – then making sure to stub in code in the main body with arguments for the new parameters making sure to return any data to a global variable.

```python
@staticmethod
def read_data_from_file(file_name: str, student_data: list):

    """..."""
    file = None
    try:
        file = open(file_name, "r")
        student_data = json.load(file)
    except Exception as e:
        print("Error: There was a problem with reading the file.")
        print("Please check that the file exists and that it is in a json format.")
        print("-- Technical Error Message -- ")
        print(e.__doc__)
        print(e.__str__())
    finally:
        # Check if a file object exists and is still open
        if file is not None and file.closed == False:
            file.close()
    return student_data
```

Figure 3: adding code from the main body

I also put in notes to remind me what was a class, function and arguments

```python
#             class                 function        Arg                          Arg
students = FileProcessor.read_data_from_file(file_name=FILE_NAME, student_data=students)
```

Figure 4: Adding code to the main body using the class

I continued this method for the variables defined in the assignments document.

- output_error_messages(message: str, error: Exception = None)
- output_menu(menu: str)
- input_menu_choice()
- output_student_courses(student_data: list)
- input_student_data(student_data: list)
- read_data_from_file(file_name: str, student_data: list):
- write_data_to_file(file_name: str, student_data: list)

```
# Presentation ------------------------------------- #
class IO:
    """..."""
    @staticmethod # print menu
    def printMenu(menu: str):...

    @staticmethod #print menu options
    def input_menu_choice():...

    @staticmethod # a general error handeling for repeated code
    def output_error_messages(message: str, error: Exception = None):...

    @staticmethod #chioce1
    def input_student_data(student_data: list):...

    @staticmethod # choice2
    def output_student_courses(student_data: list):...
```

Figure 5: the functions in der the class IO

Testing

I had to do rigorous testing that took me almost a day extra. I ended up rebuilding each section of code one function at a time and rigorously testing each step before I did the next to identify areas that were not working. I also reached out to my follow students for help.. fortunately, I was able to find my error but it did take time.

Summary

Overall, I found the method of separating code into Separations of concerns and classes and functions fairly intuitive, however I struggled with the finer nuances of changing the variables in the body of code. I feel this module would be easier if broken up in smaller bite size modules.